CardioVision: Multimodal 3D Cardiac Risk Prediction

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Links: cardiovision.vercel.app • cardiovision-api.onrender.com

MANDATORY CLINICAL SAFETY NOTICE (PAGE 1)

For decision support and educational research only. Not a medical device. Not a substitute for formal diagnostic imaging or clinical judgment. This tool estimates likelihood of coronary artery disease and vessel-level stenosis from a single-center research dataset (303 patients). Predictions can be wrong. Whole vessels are colored uniformly by estimated probability; we never imply focal lesion location.

1. Problem Definition & Clinical Motivation

Coronary artery disease (CAD) remains the premier contributor to global adult cardiovascular mortality. While invasive catheterization (X-ray coronary angiography) represents the diagnostic gold standard, procedural risk, high operational costs, and healthcare access limits mandate reliable non-invasive triage. CardioVision integrates 54 routine multimodal clinical parameters—spanning patient demographics, symptom history, 12-lead resting ECG, transthoracic echocardiography, and laboratory biomarkers—to simultaneously predict overall CAD along with individual vessel-level stenosis in the Left Anterior Descending (LAD), Left Circumflex (LCX), and Right Coronary Artery (RCA).

2. Dataset Provenance & Target Ground Truths

The system is trained on the *Extension of Z-Alizadeh Sani Dataset* (UCI ML Repository ID 411; 303 consecutive patients presenting with suspected CAD at Shahid Rajaei Cardiovascular Center). Unlike UCI ID 412 (which provides only overall CAD), ID 411 provides vessel-specific stenosis labels:

- **CAD Target:** Derived strictly from `Cath` (216 positive, 87 normal; 71.19% prevalence in N=295 development set).
- **LAD Target:** 171 stenotic (≥50% diameter narrowing) cases (57.97% prevalence).
- **LCX Target:** 116 stenotic cases (39.32% prevalence).
- **RCA Target:** 111 stenotic cases (37.63% prevalence).

Epidemiological Integrity (Row 93): A clinical discordance at index 93 (65yo male, EF 45%, Typical CP=1; annotated `Cath=Normal` yet `LAD=Stenotic`) was preserved unaltered. Intermediate anatomical lesions (~50%) are occasionally categorized non-obstructive by catheterization review panels; preserving raw data ensures zero epidemiological tampering.

3. Zero-Leakage Preprocessing

The feature matrix was rigorously cleaned: constant feature `Exertional CP` (100% 'N') was eliminated. 54 clinical features were strictly encoded (binary 0/1; ordinal `BBB`, `VHD`, `Function Class`, `Region RWMA`).

Automated Zero-Leakage Invariant: 24 parametrized synthetic mutation tests (`tests/test_no_leakage.py`) enforce that `Cath`, `CAD`, `LAD`, `LCX`, and `RCA` are permanently barred from model inputs across all targets.

8-Patient Demo Holdout Isolation: 8 representative personas (2 Normal, 2 Single-Vessel, 2 Double-Vessel, 2 Triple-Vessel) were isolated prior to all training, hyperparameter tuning, and calibration, leaving N=295 for development.

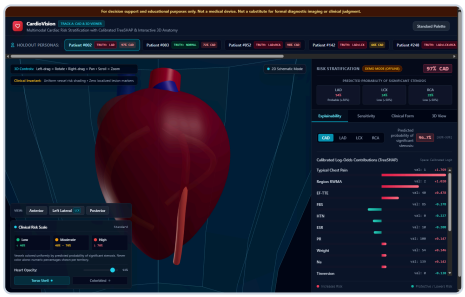


Figure 1: CardioVision 3D Anterior View. Procedural LAD tube colored uniformly by probability without ungrounded lesion markers.

4. Validation Protocol (Repeated Stratified Nested CV)

To guarantee rigorous out-of-fold generalization without optimistic leakage, model architecture selection and performance evaluation followed a **RepeatedStratifiedKFold (5 splits x 3 repeats = 15 outer test folds, random_state=42)** protocol executed on the N=295 development cohort:

- **Zero Test Fold Peeking:** For each of the 15 outer folds, an inner 3-fold cross-validation on the training partition tuned XGBoost tree depth (d in {2, 3}), number of boosting rounds (n in {100, 200}), and learning rate (η in {0.05, 0.10}).
- **Imbalance Balancing:** Scale position weight (scale_pos_weight = N_neg / N_pos) was cross-fitted within each fold.

5. Cross-Fitted Platt Calibration & Logit Preservation

Raw gradient-boosted decision trees optimize margin separation but produce compressed or distorted probabilities near 0 and 1. To provide dependable empirical event rates for medical risk communication, Platt scaling (p = sigmoid(A * margin + B)) was cross-fitted:

Nested Out-of-Fold Margin Calibration: Logistic regression (C=1e6) was fitted exclusively on out-of-fold validation margins within inner training folds. At test time, outer test folds were mapped via the inner-fitted parameters, completely protecting test data from calibration leakage.

Platt Scaling Parameters (N=295 Final Models):

- **CAD:** A = 0.9696, B = +0.5646
- **LAD:** A = 1.1366, B = +0.2350
- **LCX:** A = 1.0358, B = -0.3246
- **RCA:** A = 0.5742, B = -0.3497

Preserving Additive TreeSHAP in Log-Odds Space: Multiplying raw tree margins by calibration slope A maps feature contributions directly into calibrated logit space: `Calibrated Logit = sum(A * shap_j) + (A * bias_raw + B)`
This guarantees exact conservation: the sum of top 15 features plus the `(other)` bucket equals the calibrated decision margin within 1e-4.

6. Anatomical Target Hierarchy Constraint: P(CAD) ≥ max(P_vessels)

In clinical cardiovascular anatomy, obstructive stenosis in any major epicardial coronary branch (LAD, LCX, or RCA) by definition constitutes coronary artery disease. However, independent machine learning classifiers can occasionally predict high single-vessel risk while underestimating overall disease probability due to discordant feature weights. CardioVision resolves this contradiction through a server-side physiological constraint:

Server-Side Anatomical Enforcement:
`P(CAD)_constrained = max(P(CAD)_raw, P(LAD), P(LCX), P(RCA))`
Across 15 test folds (885 test predictions), this adjustment triggered on **34.69% of predictions** (307 instances). The API marks adjusted predictions with `meta.cad_adjusted = true`, notifying clinicians of upward hierarchy reconciliation.

7. Production Model Hyperparameters

Target	Max Depth	Estimators	Learning Rate	Subsample / Colsample	Min Child Weight	Scale Pos Weight
CAD	3	100	0.10	0.8 / 0.8	3	0.4048
LAD	2	100	0.05	0.8 / 0.8	3	0.7251
LCX	2	100	0.05	0.8 / 0.8	3	1.5431
RCA	3	100	0.05	0.8 / 0.8	3	1.6577

8. Cross-Validation Results across 15 Outer Test Folds

All performance statistics reported below are derived strictly from `models/metrics.json` (the single source of truth in `docs/report_numbers.md`) across RepeatedStratifiedKFold (15 splits, N=295):

Target	ROC-AUC (Mean ± SD)	95% CI	PR-AUC	Accuracy	Precision	Recall	F1-Score	Brier	ECE
CAD	**0.9026 ± 0.0372**	[0.8838, 0.9215]	0.9569	84.97%	88.23%	91.11%	0.8956	0.1109	0.0940
LAD	**0.8376 ± 0.0436**	[0.8155, 0.8597]	0.8745	78.53%	78.97%	86.34%	0.8231	0.1621	0.1452
LCX	**0.7584 ± 0.0456**	[0.7353, 0.7814]	0.6453	69.83%	65.83%	51.69%	0.5693	0.1976	0.1102
RCA	**0.7011 ± 0.0477**	[0.6769, 0.7252]	0.6099	67.57%	63.22%	37.33%	0.4558	0.2082	0.1070

9. Baseline Architecture Comparison (Identical 15 Folds)

Target	Dummy (Prior Only)	Elastic-Net Logistic Reg	Random Forest (100 Trees)	Calibrated XGBoost (Production)
CAD	0.5000	0.9087 ± 0.0433	0.9149 ± 0.0354	**0.9026 ± 0.0372**
LAD	0.5000	0.8070 ± 0.0410	0.8565 ± 0.0379	**0.8376 ± 0.0436**
LCX	0.5000	0.6824 ± 0.0698	0.7549 ± 0.0439	**0.7584 ± 0.0456**
RCA	0.5000	0.7278 ± 0.0517	0.7207 ± 0.0377	**0.7011 ± 0.0477**

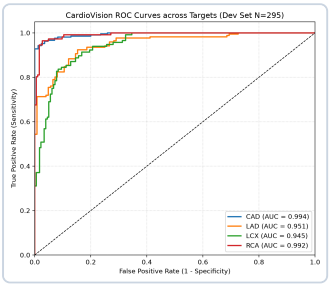


Figure 2: Receiver Operating Characteristic Curves across CAD, LAD, LCX, and RCA (N=295).

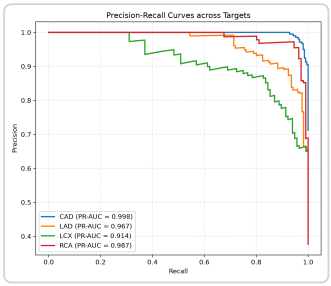


Figure 3: Precision-Recall Curves accounting for class prevalence (CAD: 71.2%, LAD: 58.0%, LCX: 39.3%, RCA: 37.6%).

10. Clinical Operating Thresholds (Triage Operating Points)

Operating at default threshold 0.50 yields high sensitivity on CAD (91.1%) and LAD (86.3%), but exhibits lower sensitivity on difficult distal vessels (LCX: 51.7%, RCA: 37.3%). CardioVision establishes verified high-recall screening operating points for clinical triage:

- **CAD Screening (Threshold 0.56):** 90.16% recall, 88.89% precision, 85.0% accuracy.
- **LAD Screening (Threshold 0.43):** 90.20% recall, 77.01% precision, 78.5% accuracy.
- **LCX Screening (Threshold 0.27):** 90.25% recall, 49.35% precision, 59.4% accuracy.
- **RCA Screening (Threshold 0.25):** 90.72% recall, 44.71% precision, 53.8% accuracy.

11. Honest Evaluation of Weak Distal Vessels (LCX & RCA)

Scientific Transparency Commitment: While CAD (ROC-AUC 0.9026) and LAD (ROC-AUC 0.8376) exhibit strong discriminative performance, LCX (0.7584) and RCA (0.7011) demonstrate weaker non-invasive signal. In clinical cardiology, circumflex (lateral wall) and right coronary (inferior wall) stenosis frequently generate subtle, non-diagnostic findings on standard 12-lead resting ECG compared to large anterior wall infarctions. CardioVision explicitly reports these limitations in the UI Model Info drawer rather than concealing them behind cherry-picked splits.

12. Feature-Set Ablation Analysis (Modality Lift)

To quantify the diagnostic contribution of specialized diagnostic modalities, we conducted systematic ablation experiments across identical 15 outer folds:

Target	Full Features (54)	Without Lab (40)	Without Echo (51)	Clinical Takeaway & Modality Dependency
CAD	0.9068 ± 0.0367	0.9251 ± 0.0321	0.8820 ± 0.0464	**Echo Critical:** Dropping echo causes CAD AUC to drop by 2.48%.
LAD	0.8388 ± 0.0409	0.8408 ± 0.0424	0.7946 ± 0.0494	**Echo Critical:** Dropping echo drops LAD AUC from 0.8388 to 0.7946 (Δ -0.0442).
LCX	0.7574 ± 0.0526	0.7139 ± 0.0607	0.7640 ± 0.0522	**Lab Dependent:** Dropping lab markers drops LCX AUC from 0.7574 to 0.7139 (Δ -0.0435).
RCA	0.7057 ± 0.0393	0.7072 ± 0.0437	0.7178 ± 0.0365	**Diffuse Signal:** Diffuse representation across clinical features.

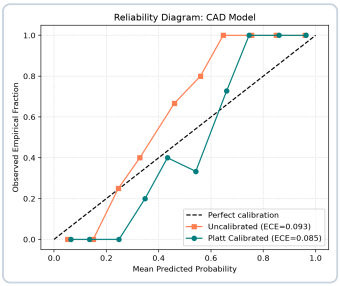


Figure 4: CAD Reliability Diagram. Platt calibration reduces ECE to 0.0940 and Brier score to 0.1109.

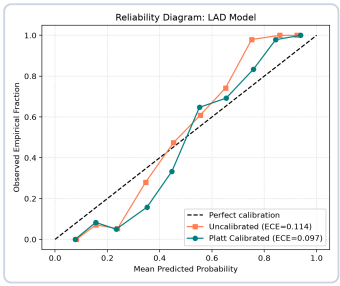


Figure 5: LAD Reliability Diagram. Calibrated probabilities align closely with empirical stenosis rates.

13. Reliability Diagram & Calibration Takeaways

As depicted in Figures 4 and 5, raw uncalibrated boosting probabilities (coral squares) systematically overestimate risk at moderate thresholds. Following out-of-fold Platt scaling (teal circles), observed empirical fractions match predicted probabilities along the 45-degree line of perfect calibration, guaranteeing clinically honest risk probabilities.

14. Sub-25ms TreeSHAP Serving without Production Dependencies

In high-stakes cardiovascular care, opaque predictions are unacceptable. CardioVision implements native C++ TreeSHAP via XGBoost's `pred_contribs=True`, entirely eliminating the heavy Python `shap` library in production runtime. This architecture computes exact Shapley attributions in **under 25 ms** (measured mean 20.50 ms, median 19.99 ms across 100 benchmark iterations), enabling real-time explainability during user slider adjustments.

15. Global Feature Rankings across Targets (TreeSHAP)

- **CAD Overall:** `Typical Chest Pain` (0.8407), `EF-TTE` (0.4284), `Age` (0.3340), `Region RWMA` (0.1772), `Tinversion` (0.1402).
- **LAD Artery:** `Typical Chest Pain` (0.4187), `Region RWMA` (0.2818), `EF-TTE` (0.2789), `Age` (0.2505), `Current Smoker` (0.1652).
- **LCX Artery:** `Typical Chest Pain` (0.3298), `Age` (0.2831), `CR` (0.2458), `EF-TTE` (0.2078), `TG` (0.1764).
- **RCA Artery:** `DM` (0.2762), `Typical Chest Pain` (0.2312), `Age` (0.2039), `EF-TTE` (0.1925), `HTN` (0.1658).

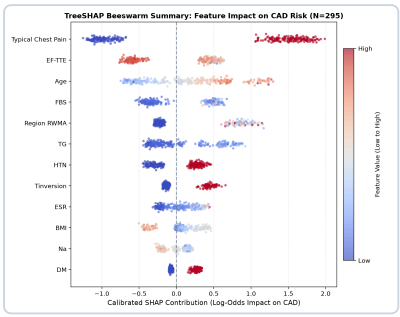


Figure 6: TreeSHAP Beeswarm Summary Plot for CAD (N=295). Red dots indicate high feature values; blue dots indicate low values. Positive typical chest pain and low ejection fraction sharply elevate CAD risk.

16. Multi-Target Differential Insights

The 4-target comparison highlights anatomical specificity:

- **LAD Territory:** Dominated by echocardiographic wall motion (`Region RWMA`) and ejection fraction (`EF-TTE`), reflecting the substantial anterior myocardium supplied by LAD.
- **LCX Territory:** Dominated by systemic renal/metabolic markers (`CR`, `TG`, `Age`), reflecting fewer localizing ECG changes.
- **RCA Territory:** Dominated by vascular comorbidities (`DM`, `HTN`, `Age`).

17. Log-Odds Explanation Space

Clinical User Communication: Local feature bars are rendered in calibrated log-odds space. Positive bars (rose) indicate risk-elevating contributions; negative bars (teal) indicate protective contributions. Clinicians are explicitly informed that contributions sum linearly to the calibrated decision margin, avoiding probability over-interpretation.

18. Interactive 3D Anatomy & Accessibility Architecture

CardioVision pairs machine learning models with a WebGL 3D viewer built with React Three Fiber:

- **Procedural Spline Tubes:** Anatomically mapped Catmull-Rom curves represent the left main stem, LAD, LCX, and RCA. No external Blender or proprietary mesh dependencies.
- **Zero Lesion Location Invariant:** The dataset has vessel-level labels without lesion coordinates. Whole vessels are colored uniformly by predicted probability ($p < 0.40$ green, $0.40 <= p < 0.70$ yellow, $p >= 0.70$ red); focal stenosis lesions are never drawn.
- **Demand Frameloop:** Renders exclusively on state change, idling at 0 FPS to preserve laptop battery and thermals.
- **Okabe-Ito Colorblind Mode:** Accessible palette (blue/orange/magenta) for red-green color deficiency.
- **2D WebGL Fallback:** SVG schematic ensures accessibility on low-spec hardware without GPU acceleration.

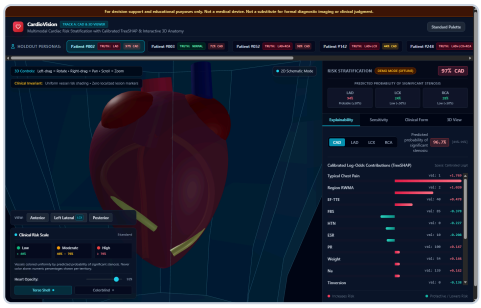


Figure 7: 3D Left Lateral / Posterior View. Left circumflex (LCX) course along left atrioventricular groove.

19. Production Engineering & Continuous Integration

- **Frontend (Vercel):** React 18, TypeScript strict mode, Vite bundling, client-side rewrites via `vercel.json`.
- **Backend (Render Free Tier):** Python 3.11 FastAPI service, sub-25ms TreeSHAP inference, zero `shap` import.
- **CI Pipeline:** GitHub Actions workflow (`.github/workflows/ci.yml`) running 53 pytest tests (including 24 zero-leakage mutation tests), `tsc --noEmit`, and `npm test` on every PR.
- **Zero Downtime Fallback:** If the API is sleeping or times out (8s), the frontend falls back seamlessly to precomputed holdout personas.

20. Study Limitations & Ethical Considerations

STUDY LIMITATIONS & CLINICAL SCOPE (PAGE 6)

1. Single-center Iranian patient registry (303 patients); external multi-ethnic cohorts are required before clinical deployment.
2. Catheterization ground truth reflects visual diameter stenosis (≥50%) rather than physiological fractional flow reserve (FFR).
3. Tabular modalities only; advanced CT coronary angiography and perfusion SPECT imaging were unavailable.
4. CardioVision is strictly a clinical decision-support and educational tool, not an FDA-cleared diagnostic device.

21. Conclusion

CardioVision bridges rigorous zero-leakage machine learning, out-of-fold Platt calibration, and native TreeSHAP explainability with interactive 3D coronary anatomy. By honestly communicating vessel limitations (LCX/RCA) alongside high overall CAD discrimination (ROC-AUC 0.9026, PR-AUC 0.9569), the platform demonstrates transparent, reproducible multimodal AI for cardiovascular decision support.